

Q. 1 (Destination unknown). Passengers arrive at a bus stop as a Poisson process with rate λ . Each passenger is headed towards one of two destinations A and B with probabilities p and $1 - p$, respectively. Buses to A arrive as a Poisson process with rate μ_A , while buses to B arrive as a Poisson process with rate μ_B . The arrivals of the buses and passengers and the destination of each passenger are all independent.

- How many people on average get on the first bus to A ?
- How many people on average get on the first bus to arrive at the stop?
- How many people on average remain at the stop after the first bus leaves?

Q. 2 (Art sale). You have a priceless Van Gogh painting that you inherited from your grandfather. Unfortunately, a frozen yogurt addiction has rendered you short of funds, and you are compelled to sell. Potential buyers arrive to bid on the painting according to a Poisson process with rate of λ buyers per day. The k th buyer bids $\$X_k$ for the painting. Assume that $\{X_k\}$ are independent exponential random variables with parameter μ that are independent of the arrival times of the potential buyers.

You would like to get a high price for the painting, so you may be willing to wait until a large bid comes in. Meanwhile, however, the painting must be stored in a temperature-controlled vault that is insured against theft. The storage costs run to $\$C$ per day.

- What is your expected net profit if you sell to the first buyer who comes along?
Could it be more profitable to junk the painting than to sell to the first buyer?

Now consider the following strategy for selecting a buyer for your painting: fix a number $y \geq 0$, and sell to the first buyer who bids at least $\$y$ for the painting.

- What is the expected amount of time until you sell the painting?
- What is the expected price you get for the painting?

Hint: show that if you have a sequence of i.i.d. random variables $\{X_k\}$, and you wait until the first trial T that $X_T \geq y$, then $\mathbf{E}(X_T) = \mathbf{E}(X_1 | X_1 \geq y)$.

- What is the value of y that will get you the largest expected profit? Does this strategy yield a larger expected profit than if you were to sell to the first buyer?

Q. 3 (Casino). In a Las Vegas casino, the average gambler starts with about $\$10,000$ in his pocket. He is playing against the house, which has $\$10,000,000$ in the house safe. The gambler keeps gambling (with a stake of $\$1$ in each game) until either he is bankrupt, or the house is bankrupt. You may assume that the game is fair, i.e., the probability of winning or losing in each round is $\frac{1}{2}$.

- What is the probability that the gambler will go bankrupt?
- How many games does the gambler have to play, on average, before either he or the house goes bankrupt?
- How do the answers to a. and b. change if the gambler stakes \$1000 in each game?

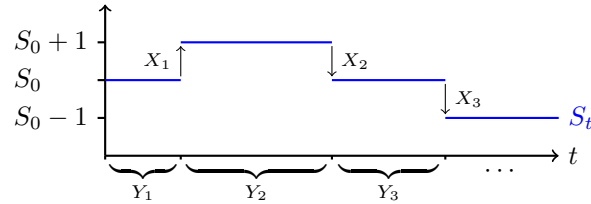
Q. 4 (Bernoulli processes revisited). Let (X_k) be a Bernoulli process, i.e., $X_1, X_2, \dots$ are i.i.d. Bernoulli variables with $\mathbf{P}\{X_k = 1\} = p$ and $\mathbf{P}\{X_k = 0\} = 1 - p$. Denote by S_n the number of successes in n trials, that is,

$$S_n = X_1 + X_2 + \dots + X_n,$$

and denote by T_k the time at which the k th success occurs. We already studied such processes in great detail earlier in this course. On the other hand, we can clearly view (S_n) as a special case of a random walk, albeit not a simple or symmetric one.

- Explain why T_k is a hitting time of the random walk (S_n) .
- Use first-step analysis to compute $\mathbf{E}(T_k)$.
- How does the answer in b. match what you already know about Bernoulli processes?

Q. 5 (Continuous time random walk). The random walks we are investigating take steps in discrete time (for example, once a second). One could also investigate random walks where steps can be taken at any time. One way to do this is as follows. Let $X_1, X_2, \dots$ be i.i.d. step sizes, independent of the initial position S_0 . Assume that $\mathbf{P}\{X_i = +1\} = \mathbf{P}\{X_i = -1\} = \frac{1}{2}$. Let $Y_1, Y_2, \dots$ be i.i.d. exponential random variables with rate λ , independent of the step sizes and the initial position. The random walk stays at S_0 for Y_1 seconds, then jumps by X_1 ; then stays put for Y_2 seconds, then jumps by X_2 ; etc. This is illustrated in the following figure:



Denote by S_t the position of the random walk at any time $t \geq 0$. Let $a \leq S_0 \leq b$.

- What is the probability that the continuous time random walk hits b before a ?
- How long does it take, on average, before the process $\{S_t\}$ hits either a or b ?
- What is the distribution of the number of steps taken by time t ?